Code No:138DU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, July - 2023
NEURAL NETWORKS AND DEEP LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Write the difference between supervised and unsupervised learning.	[2]
b)	What are the types of sigmoid function? Explain.	[3]
c)	What is self-organization?	[2]
d)	Show the structure of Kohonen's SOFM.	[3]
e)	What is gradient descent?	[2]
f)	Describe Back Propagation in the learning.	[3]
g)	Define semi supervised learning.	[2]
h)	Explain constrained optimization.	[3]
i)	List any four computer vision application.	[2]
j)	Explain Meta algorithms.	[3]

PART – B

(50 Marks)

2.a) Describe perceptron learning rule and delta learning rule.
b) Explain the algorithm of discrete Hopfield network with its architecture. [5+5]

OR

3.a) Describe winner-take-all learning rule and outstar learning rule.
b) Distinguish between Auto associative Memory and Hetero Associative Memory. [5+5]

4.a) Provide the training algorithm for MAXNET.
b) Provide the training algorithm for Hamming Network. [5+5]

OR

5. Consider a Kohonen network with two cluster units and five input units. The weigh vectors for the cluster units are
w1 = [0.1, 0.3, 0.5, 0.7, 0.9]
and
w2 = [0.9, 0.7, 0.5, 0.3, 0.1]
Use the square of the Euclidean distance to find the winning cluster unit for the input pattern. [5+5]

6.a) Explain sigmoid units for Bernoulli Output Distributions.
b) Justify the importance of rectified linear units in Hidden units. [5+5]

OR

7.a) Give an example to explain the concept of back propagation algorithm.
b) Describe Logistic Sigmoid and Hyperbolic Tangent. [5+5]

8.a) Explain L-1 regularization with example.
b) Explain various data augmentation methods in detail. [5+5]

OR

9.a) List out the advantages of ensemble methods.
b) Explain bagging with suitable example. [5+5]

10.a) Define optimization. Explain the challenges of neural network optimization.
b) Explain the applications of large scale deep learning methods with suitable example. [5+5]

OR

11.a) Explain how do you optimize a neural network with suitable example?
b) Discuss in detail about the adaptive learning rates. [5+5]

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